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REVIEW

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Smart healthcare: the role of AI, robotics, and NLP in advancing telemedicine and remote patient monitoring

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Abstract

Background The integration of artificial intelligence (AI), robotics, and natural language processing (NLP) is transforming healthcare delivery through telemedicine and remote patient monitoring (RPM). These technologies are increasingly being adopted to enhance diagnostic accuracy, patient engagement, and healthcare accessibility.

Main body This review examines the application of AI, robotics, and NLP across various medical domains including cardiology, diabetes, gynaecology, dermatology, and general RPM.

- AI facilitates precise diagnosis, predictive analytics, and personalized medicine.
- NLP enables efficient processing of medical records, symptom triage, and virtual health assistance.
- Robotics supports minimally invasive surgery, remote interventions, and routine clinical tasks.

Case studies highlight successful implementations and measurable outcomes demonstrating improved efficiency and patient outcomes. However, several challenges persist, such as concerns over data privacy, high implementation costs, lack of clinician training, and ethical considerations. Furthermore, integration of these advanced tools into cohesive telemedicine systems remains limited. Regulatory frameworks addressing safety, accountability, and standardization are still in development.

Conclusions Addressing existing technological, ethical, and policy gaps is essential to fully realize the potential of intelligent healthcare systems. Strengthening regulatory standards, ensuring interoperability, and enhancing clinician preparedness will be key to achieving effective, equitable, and safe patient care in the digital era.

Keywords Remote consultation, Artificial intelligence, Robotics, Natural language processing, Large language models, Telemedicine, Digital health, Remote patient monitoring

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Introduction

Digital technologies have profoundly reshaped healthcare delivery, particularly through remote healthcare, which broadly refers to the use of digital tools to deliver medical services across distances. Remote healthcare encompasses telemedicine (real-time virtual consultations), telehealth (a wider set of digital health services including education and prevention), and remote patient monitoring (RPM) (continuous data collection through connected devices). These approaches have expanded access to medical services, especially for underserved and geographically remote populations, while reducing the need for frequent in-person visits [1–4]. Within this context, artificial intelligence (AI) has emerged as a critical enabler. AI techniques—including machine learning (ML), natural language processing (NLP), and computer vision—enhance diagnostic accuracy, support real-time clinical decision-making, and facilitate proactive disease management [5–7]. When combined with wearable sensors, mobile health platforms, and the Internet of Things (IoT), AI enables continuous patient monitoring and personalized interventions, thereby strengthening patient–provider connectivity [3, 4, 8].

The integration of these technologies has contributed to the personalization of patient care, shifting from a centralized, institution-based approach to more individualized healthcare management systems [5, 10]. In an era marked by rapid cultural and technological change, continuous evaluation of digital innovations is essential to ensure their effectiveness and efficiency in enhancing human health and well-being.

The World Health Organization (WHO) defines eHealth as "the economical and secure application of ICTs in support of health and health-related fields, including healthcare services, health surveillance, health literature, and health education, knowledge, and research" [9].

The COVID-19 pandemic underscored the critical need to minimize physical contact through remote healthcare solutions, while simultaneously alleviating the burden on healthcare systems. AI-driven remote healthcare plays a vital role in ensuring continuity of care by enabling the monitoring, diagnosis, and management of patients at a distance [11, 12]. Remote healthcare is both convenient and cost-effective, as it significantly reduces the need for in-person consultations and hospital admissions. AI-powered platforms facilitate proactive patient engagement by enabling individuals to manage their health, follow personalized care plans, and maintain effective communication with healthcare professionals [13]. These intelligent systems can also alert patients to potential health issues, track clinical parameters, and support timely medical interventions. The evolution of AI in healthcare is presented in Table 1.

This paper demonstrates that the integration of AI, NLP, and robotics in telemedicine enables a highly synergistic approach to patient care. AI interprets complex clinical data, NLP translates patient communication and medical records into actionable insights, and robotics facilitates real-time physical interventions. Together, these technologies enhance efficiency and accessibility, allowing clinicians to remotely monitor, diagnose, and treat patients, thereby extending high-quality care to underserved or rural areas. This combination also improves decision-making by creating a closed-loop system in which AI-driven analytics, NLP-derived patient insights, and robotic execution enable continuous monitoring, early intervention, and personalized care. For example, during a teleconsultation, NLP chatbots can collect patient symptoms, AI can evaluate risks and

Table 1 AI evolution in healthcare over the years

Era	Description
Early Foundations (1950–1970 s)	The term *"Artificial Intelligence"* was coined at the 1956 Dartmouth Workshop. Focused on symbolic reasoning and rule-based systems to mimic human cognition. In medicine, this led to expert systems like MYCIN (early 1970s, Stanford), which diagnosed bacterial infections using rule-based logic. Despite its innovation, MYCIN wasn't used clinically due to limitations in computing power and integration challenges [51].
Expert Systems Era (1970–1980 s)	Expansion of rule-based expert systems such as INTERNIST-I and QMR, which encoded physician diagnostic reasoning. These systems showed promise in labs but struggled in real-world use due to the *knowledge acquisition bottleneck*—difficulty in accurately programming complex medical expertise.
Stagnation and Slow Progress (1990s)	Marked by limited progress in medical AI due to restricted computing power, lack of large datasets, and challenges in clinical integration. These limitations dampened enthusiasm and investment in the field [7].
Rise of Machine Learning (2000s)	Digitization of health data and EHR adoption enabled machine learning to analyze large datasets. This era marked a shift from rule-based to data-driven approaches, leading to improvements in predictive analytics and diagnostic support [6].
Deep Learning and Big Data (2010s–Present)	Advancements in deep learning (multi-layer neural networks) and availability of big data transformed diagnostics. AI systems began achieving expert-level performance, especially in medical imaging (e.g., radiographs, MRIs). Dermatology saw enhanced skin cancer detection [20].
AI Integration in Clinical Practice (2020s–Ongoing)	AI is increasingly embedded in clinical workflows: virtual health assistants, surgical planning aids, patient monitoring, and platforms like Doctronic offer real-time consultations. AI also helps manage surgical wait times and supports personalized prehabilitation programs, improving patient care outcomes [8].

suggest treatment plans, and robotic systems can administer therapies or assist in remote procedures.

This review critically examines the transformative impact of AI on remote healthcare, with a particular emphasis on enhancing patient engagement and connectivity. It explores recent applications of AI in the delivery of healthcare services at a distance, highlighting how these technologies facilitate improved care coordination and communication between patients and providers. Additionally, the role and significance of large language models is explored. Therefore, the advantage and contribution of robotics in healthcare is explored. Consequently, the paper provides real time case studies of AI in healthcare. Finally, the conclusion offers insights into future directions and emerging innovations that are likely to shape the evolution of AI-driven remote healthcare.

AI in health monitoring

Artificial intelligence (AI) refers to the simulation of human cognitive abilities—such as planning, reasoning, and decision-making—by machines across various domains. A prominent subset of AI, machine learning (ML), leverages algorithms and data to emulate human learning processes [14]. In healthcare, AI and ML have become instrumental in enhancing patient care through improved diagnostic accuracy, efficiency, and speed [15].

One of the most widely studied applications of AI is medical imaging analysis, where algorithms outperform or complement human radiologists in detecting abnormalities on X-rays, MRIs, CT scans, and ultrasounds [16]. Comparative studies have shown that deep learning models can achieve diagnostic accuracies comparable to, and sometimes exceeding, expert clinicians—particularly in oncology and cardiology. However, their performance often depends on the size and quality of training datasets, raising concerns about generalizability across populations [17].

Beyond imaging, AI can process diverse patient data—such as bio signals (ECG, EEG, EMG), vital signs, electronic health records (EHRs), demographics, and lab results—to support predictive analytics and personalized interventions [18]. This is particularly relevant for chronic and cardiovascular conditions, where early detection is essential [19]. Cardiology, for instance, is experiencing a paradigm shift as AI-based models predict arrhythmias or heart failure exacerbations earlier than conventional monitoring tools. At the same time, critics note that black-box AI models limit clinical interpretability, creating challenges for physician acceptance and regulatory approval [20]. The role of AI is depicted in Fig. 1.

AI's integration into remote healthcare management is reshaping service delivery, especially in underserved regions. However, its role extends beyond technological novelty—it must be assessed critically in terms of scalability, cost-effectiveness, and ethical deployment. Below, key applications are outlined along with their opportunities and limitations.

Remote patient monitoring (RPM)

AI-powered systems track vital signs in real time (heart rate, blood pressure, glucose, oxygen saturation) and alert providers when parameters deviate from normal ranges. Comparative studies show improved early detection of complications compared to conventional monitoring. Still, false positives can generate alert fatigue, while data privacy and secure transmission remain major challenges.

Telemedicine and virtual consultations

AI chatbots and virtual assistants triage queries, schedule appointments, and provide follow-up care. Evidence suggests these tools reduce clinician workload, but their accuracy in complex cases is limited, raising risks of misdiagnosis if used without physician oversight.

AI for chronic disease management

AI enables personalized recommendations for diabetes, heart disease, and hypertension management, reducing frequent hospital visits. Yet, long-term adherence to AI-driven recommendations is inconsistent, and there is limited evidence from large-scale randomized trials confirming clinical effectiveness.

AI in diagnostics

Image-based AI tools accelerate early disease detection in oncology, neurology, and cardiovascular care. While they reduce diagnostic delays, there is controversy over bias in training datasets, which may underrepresent minority groups and yield unequal outcomes.

Predictive analytics for disease prevention

By analyzing lifestyle, genetic, and clinical data, AI predicts disease risk and guides preventive interventions. However, predictive accuracy varies across populations, and ethical concerns arise regarding overdiagnosis and the psychological impact of labeling patients as "high risk."

AI-based medication management

Medication adherence apps and wearable sensors reduce missed doses and adverse events. Yet, studies highlight low patient engagement over time, suggesting usability and digital literacy remain barriers.

AI in mental health monitoring

Wearable and voice-based AI systems monitor stress, depression, and anxiety. While promising for early crisis

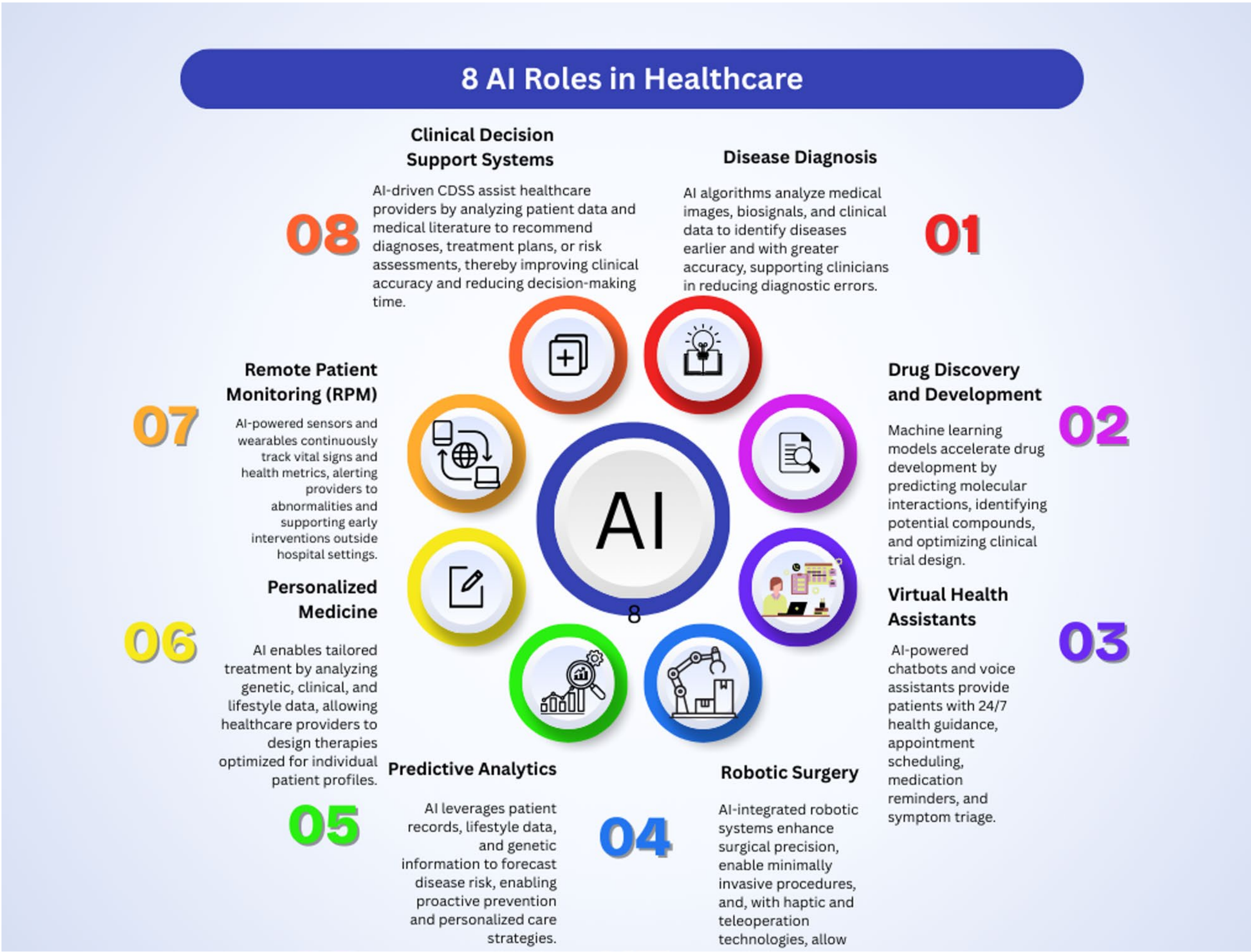

Fig. 1 Role of AI in health monitoring and telemedicine

detection, the field faces controversies about data sensitivity and the ethical use of personal behavioral data (e.g., from social media).

AI for population health management
Population-level AI analytics help predict disease outbreaks and allocate resources. These approaches can complement traditional epidemiology but raise debates about data ownership, particularly when private tech firms manage health data.

Natural language processing (NLP) for medical records
NLP extracts insights from unstructured clinical notes, supporting informed decision-making. However, its performance is often limited by language diversity and documentation quality, reducing effectiveness in multilingual or resource-limited settings.

Wearable AI-powered health devices
Wearables like smartwatches integrate AI for continuous monitoring. They promote patient empowerment but also generate massive unvalidated data, overwhelming providers and raising uncertainty about long-term clinical impact.

AI in health monitoring offers remarkable opportunities for early detection, continuous care, and population health management. Yet, limitations such as dataset bias, lack of interpretability, privacy risks, and uneven evidence across clinical contexts temper the optimism. The literature highlights a clear need for standardized validation studies, ethical frameworks, and regulatory guidance to ensure that AI systems not only function effectively but also integrate responsibly into healthcare ecosystems.

Telemedicine and remote patient monitoring
The delivery of healthcare services through audio-visual communication—has expanded access for remote and underserved populations, enabling consultations, monitoring, and treatment without physical presence [21–23]. A critical component is the secure transmission of health data, often automated through digital platforms and smart devices.

Within this landscape, Remote Patient Monitoring (RPM) has emerged as a central tool. Evidence shows that RPM improves clinical outcomes, reduces hospital readmissions, and enhances patient and provider satisfaction [24, 25]. For example, Motolese et al. [24] demonstrated telemedicine's effectiveness for neurological patients during COVID-19, even in populations with limited digital literacy [26]. Similarly, home-based blood pressure monitoring and reporting systems have proven more effective for long-term hypertension management compared to routine clinic visits [27, 28]. These findings suggest that telemedicine and RPM not only expand reach but can also improve disease control when properly integrated into care pathways.

However, the literature is not uniformly positive. Several studies highlight barriers to sustained adoption. These include digital literacy gaps among elderly patients, limited broadband access in rural regions, and concerns over data privacy and security [29, 30]. While RPM was widely adopted during the pandemic for post-discharge COVID-19 patients, follow-up research indicates challenges in scalability and sustainability, particularly when emergency funding and temporary infrastructures were withdrawn [29, 31].

Another area of debate is cost-effectiveness. While RPM may reduce hospital readmissions and emergency visits, initial investments in digital infrastructure, training, and device distribution can be high. Some studies suggest long-term savings, while others caution that benefits vary significantly depending on the condition monitored, patient adherence, and the healthcare system's reimbursement models [32, 33].

Patient and provider perceptions also influence outcomes. Many patients report greater convenience and empowerment through RPM, yet others express concerns about impersonal care and lack of face-to-face interaction. Providers, meanwhile, value improved monitoring but cite alert fatigue from excessive system notifications and worry about increased liability when continuous monitoring is expected without adequate staffing [31, 33].

Overall, telemedicine and RPM represent promising innovations with demonstrated benefits in chronic disease management, emergency response, and rural healthcare delivery. Yet, their long-term impact depends on addressing critical barriers around equitable access, cost sustainability, user acceptance, and regulatory frameworks. A more systematic evaluation of these factors is needed before telemedicine can fully transition from a pandemic-driven solution to an integral component of routine healthcare delivery.

Large language models in medicine

Large language models (LLMs), powered by advanced AI techniques, are designed to generate human-like language by learning from massive datasets, including internet-based and domain-specific content [1, 2]. These models perform a wide range of language-related tasks, such as summarization, translation, paraphrasing, storytelling, and question answering. The release of OpenAI's ChatGPT in November 2022 marked a major milestone in public engagement with LLMs, demonstrating their applicability in medicine, education, research, and clinical decision support. In medical contexts, LLMs have been applied to generate educational content, assist in literature reviews, and support preliminary diagnostic reasoning [34, 35].

Despite their promise, LLMs raise significant accountability, transparency, and reliability concerns. A key challenge is the phenomenon of hallucination, where models generate plausible but incorrect or fabricated information. In clinical settings, such errors could lead to harmful or even fatal outcomes [34, 36]. LLMs are also prone to replicating societal biases present in their training data, which may reinforce diagnostic disparities, particularly across gender, ethnicity, or socioeconomic groups. While safety mechanisms—including model fine-tuning, prompt engineering, and post-processing validation—have been developed to mitigate bias and hallucination, their effectiveness is still inconsistent [35, 36].

Another critical debate revolves around regulatory and medico-legal frameworks for LLM deployment in healthcare. Regulatory authorities are increasingly focused on establishing guidelines for model validation, data provenance, transparency, and accountability, particularly in high-stakes clinical environments [36, 37]. The absence of clear legal and ethical guidance raises questions about liability when AI-generated outputs influence patient care. Who is responsible if a clinician follows LLM guidance that leads to patient harm: the developer, the healthcare provider, or the institution? Current literature underscores that clarifying liability, establishing robust monitoring, and integrating human oversight are essential prerequisites for safe adoption [36, 37].

Emerging LLMs trained on medical and clinical datasets show promise in addressing some of these challenges by reducing hallucination, improving accuracy, and aligning outputs with established medical knowledge [35]. However, wide-scale adoption requires careful consideration of ethical implications, continual model auditing, and adherence to regulatory standards, as well as robust integration into existing clinical workflows to ensure that AI augments rather than replaces human judgment.

While LLMs represent a transformative tool for medical knowledge dissemination, education, and clinical support, their deployment must be accompanied by critical oversight, evidence-based validation, and governance frameworks to prevent errors, bias, and legal complications in healthcare settings [36, 37]. Key challenges, ongoing debates, and mitigation strategies for LLM deployment in healthcare are summarized in Table 2.

Robotics and healthcare

The integration of robotics into healthcare is transforming medical practice, driven by advances in AI, miniaturization, and computational power [36, 37]. Robotic systems have been most impactful in surgery, enabling precise manipulation of instruments through minimally invasive techniques. Using high-definition 3D imaging and computer-guided controls, robotic-assisted surgery improves accuracy, reduces blood loss, minimizes incisions, and shortens recovery times compared to traditional approaches. The da Vinci Surgical System, FDA-approved in 2000, has been utilized in over six million surgeries worldwide, demonstrating the scalability and clinical effectiveness of robotic interventions [36, 38]. Similar systems have been applied across specialties such as urology, gynecology, and cardiothoracic surgery, showing consistent reductions in postoperative complications and hospital stays.

Beyond surgery, robotics is advancing diagnostics and treatment. Innovations such as microrobots and robotic endoscopic capsules enable targeted drug delivery, biopsies, cauterization, and gastrointestinal monitoring, offering minimally invasive alternatives to conventional procedures [37]. Robotics also supports routine clinical tasks including patient monitoring, data entry, blood sampling, and logistics management, which can enhance efficiency in hospitals and clinics. Emerging nanorobots, capable of binding to specific pathogens, present novel approaches for targeted infection control, potentially reducing reliance on broad-spectrum antibiotics [37].

The intersection of robotics with remote monitoring and telemedicine is becoming increasingly relevant. Advances in haptic technologies, combined with low-latency 5G networks, allow surgeons to perform procedures remotely while receiving real-time tactile feedback [10, 39]. This capability not only extends specialized surgical expertise to underserved or rural regions but also integrates robotics into broader telehealth ecosystems, complementing Remote Patient Monitoring (RPM) initiatives.

However, several emerging challenges must be addressed for widespread adoption. Robotic systems involve substantial cost barriers, including purchase, maintenance, and software updates, which may limit access in low-resource settings. Effective deployment requires extensive training for surgeons and clinical staff, as procedural complexity increases with automation. Ethical concerns also arise, including the degree of human oversight in autonomous or semi-autonomous interventions, patient consent for robotic procedures, and accountability in cases of adverse events [36, 39].

Robotics offers transformative potential for surgery, diagnostics, and remote healthcare integration. Yet, successful implementation requires balancing clinical benefits with cost, training, ethical, and regulatory considerations, ensuring that robotic solutions enhance rather than replace human clinical judgment.

Case studies of AI in health diagnosis

Case studies of telemedicine and AI in healthcare highlight significant improvements in patient care and accessibility. For example, India's eSanjeevani platform has facilitated millions of remote consultations, bridging rural healthcare gaps. In the U.S., the Mayo Clinic uses AI-driven algorithms to predict patient deterioration, enabling early intervention. Babylon Health in the UK employs AI chatbots for preliminary assessments, reducing clinician workload. Another case, Stanford's use of

Table 2 Key challenges, ongoing debates, and mitigation strategies for LLM deployment in healthcare

Challenge / Debate	Description	Potential Mitigation / Strategy	References
Hallucination	LLMs may generate plausible but incorrect or fabricated content.	Fine-tuning on medical datasets, prompt engineering, post-processing validation, human oversight.	[34, 52]
Bias and Equity	Models replicate societal biases, affecting diagnostic accuracy across gender, ethnicity, and socioeconomic groups.	Diverse training datasets, bias auditing, model explainability, clinical validation.	[35, 52]
Regulatory Compliance	Lack of established guidelines for clinical validation, safety, and deployment.	Develop regulatory frameworks, model certification, continuous monitoring, compliance with data privacy laws.	[52, 53]
Medico-Legal Liability	Unclear responsibility when AI-generated outputs influence clinical decisions and patient outcomes.	Establish institutional protocols, human-in-the-loop decision-making, clear legal guidelines.	[52, 53]
Safety & Reliability	Risk of inaccurate or unsafe recommendations in high-stakes clinical environments.	Evidence-based model evaluation, scenario testing, integration into clinical workflows with oversight.	[35, 52]
Adoption & Trust	Clinician and patient skepticism limits utilization.	Training and education, transparency about model limitations, decision support rather than replacement.	[35, 53]

Table 3 Key findings from case studies on AI in remote health monitoring

Application Area	Case Study / System	AI Techniques Used	Key Findings	Outcomes	Reference
Cardiology	TeleHealth for Heart Failure Monitoring	Machine Learning (ML), Predictive Analytics	Early detection of decompensation events in heart failure patients using real-time monitoring	Reduced hospital readmissions and improved patient self-management	Kang et al., 2022 [46]
Diabetes	AI-Based Glucose Monitoring (e.g., FreeStyle Libre)	Decision Support Systems, Deep Learning	Continuous glucose data interpretation to guide insulin dosage and meal planning	Improved glycemic control and patient adherence	Alavi et al., 2021 [47]
Dermatology	SkinVision App	Convolutional Neural Networks (CNN)	AI model detects suspicious skin lesions via mobile image uploads	Increased early detection rates of melanoma and other skin cancers	Thissen et al., 2020 [48]
Gynaecology / Maternity	AI in Antenatal Monitoring in Malawi	Computer Vision, Risk Stratification Algorithms	AI-powered foetal and maternal vital sign monitoring helped identify complications early	Significant reduction in stillbirths and neonatal mortality	Chiweza et al., 2024 [49]
General Remote Monitoring	Cloud-based AI for RPM (India)	Cloud-AI Integration, ML for predictive alerts	Integrated real-time health monitoring with cloud computing for various chronic illnesses	Enabled proactive interventions, reduced emergency events	Benadikar, 2021 [50]

deep learning to interpret chest X-rays, has matched radiologist-level accuracy. These examples demonstrate how combining AI with telemedicine enhances diagnostic accuracy, streamlines workflows, and expands access to quality healthcare across diverse populations. Additionally, certain case studies are presented in Table 3 in this review—covering cardiology, diabetes, gynaecology, dermatology, and general remote patient monitoring. These domains are selected through a systematic and rationale-driven process. First, these domains were identified based on prevalence, clinical impact, and the demonstrated integration of AI, NLP, and robotics in healthcare literature. Peer-reviewed journals, conference proceedings, and authoritative reports were screened for applications where multiple technologies converged to deliver measurable improvements in diagnosis, treatment, or patient engagement. The inclusion criteria were; Relevance: studies must clearly implement AI, NLP, or robotics in a healthcare context; Diversity: Domains were chosen to represent a range of specialties (chronic disease, surgery, remote monitoring, and diagnostics); Impact and Evidence: Preference was given to studies reporting quantitative outcomes, clinical benefits, or novel technological integration; Recency: Priority was given to research published within the last 5–7 years to capture contemporary trends in smart healthcare. Furthermore, several prominent case studies are presented as follows:

AI-enabled remote patient monitoring (RPM) architectures

Shaik et al. (2023) conducted a comprehensive review of AI-integrated RPM systems, highlighting the use of Internet of Things (IoT) devices, cloud computing, and machine learning algorithms to monitor patients' vital signs and activities remotely. The study emphasizes the role of AI in early detection of health deterioration, personalized monitoring through federated learning, and adaptive learning of patient behavior using reinforcement learning techniques. These advancements have transformed healthcare monitoring by enabling non-invasive, continuous, and real-time patient assessment [38].

Indigenous AI-based RPM devices: Dozee.ai and Qure.ai

Krishnaveni et al. (2023) presented a case study on the implementation of AI-driven RPM devices, Dozee.ai and Qure.ai, in India. These devices have been instrumental in monitoring patients remotely, especially during the COVID-19 pandemic. Dozee.ai utilizes contactless sensors to track vital parameters, while Qure.ai employs AI algorithms to interpret medical imaging. The study underscores the effectiveness of these indigenous technologies in providing timely and cost-effective healthcare solutions, despite challenges related to data privacy and device accuracy [39].

Cloud-based AI and machine learning techniques in personalized healthcare

Benadikar et al. (2021) evaluated the effectiveness of cloud-based AI and machine learning techniques in personalized healthcare and RPM. The study analyzed applications such as predictive analytics, natural language processing, and wearable device integration. Case studies in cardiovascular disease, diabetes, mental health, and post-operative care demonstrated improvements in treatment plan optimization, risk stratification, and patient engagement. The research highlights the potential of cloud-based AI solutions in enhancing clinical outcomes and healthcare delivery efficiency [40].

AI monitoring in maternity care

A practical application of AI in remote health monitoring is observed at the Area 25 health center in Lilongwe, Malawi. The center implemented AI-enabled fetal monitoring software to track babies' vital signs during labor. This technology facilitated real-time detection of fetal distress, leading to timely interventions. Since its adoption, the clinic reported an 82% reduction in stillbirths and neonatal deaths, showcasing the profound impact of AI on maternal and neonatal healthcare outcomes [41].

AI-enabled remote ECG monitoring for asymptomatic myocardial ischemia
A study by Lv et al. (2021) evaluated an AI-driven remote electrocardiogram (ECG) monitoring system for patients with coronary heart disease. The system utilized AI algorithms to analyze ECG data, effectively identifying asymptomatic myocardial ischemia. The integration of AI allowed for continuous remote monitoring, facilitating early detection and timely intervention, thereby improving patient outcomes [42].

Machine learning-based remote monitoring framework for diabetes prediction
Ramesh et al. (2021) proposed a remote healthcare monitoring framework employing machine learning algorithms to predict diabetes risk. The system integrated data from personal health devices and smartphones, utilizing a support vector machine model trained on the Pima Indian Diabetes Database. The model achieved an accuracy of 83.2%, sensitivity of 87.2%, and specificity of 79%, demonstrating its potential in facilitating early detection and management of diabetes through remote monitoring [43].

AI-based digital health technology for non-melanoma skin cancer detection
In the DERM-003 study, researchers assessed the effectiveness of an AI-based digital health technology in identifying non-melanoma skin cancer and other skin lesions. The AI system analyzed images of skin lesions, demonstrating high accuracy in detecting basal cell carcinoma and squamous cell carcinoma. This technology enables primary care physicians and nurse practitioners to perform accurate remote assessments, enhancing early detection and treatment of skin cancers [44].

AI-powered remote monitoring in obstetrics and gynaecology
A comprehensive review by Sharma et al. (2024) highlighted the transformative role of AI in obstetrics and gynaecology. AI-powered systems analyze data from cardiotocography, foetal ultrasounds, and maternal physiological parameters to monitor foetal health remotely. These systems can detect early signs of foetal distress, intrauterine growth restriction, and other complications. Additionally, AI aids in monitoring maternal health by analyzing vital signs and laboratory results, facilitating timely interventions and improving outcomes in high-risk pregnancies [45].

Conclusion
The integration of artificial intelligence into healthcare is transforming the way medical services are delivered, diagnosed, and managed. From AI-powered telemedicine platforms and remote health monitoring systems to natural language processing tools that streamline clinical documentation, and robotics that enhance surgical precision and patient care, these technologies are collectively reshaping the healthcare landscape. Together, they offer the potential to improve patient outcomes, increase efficiency, and expand access to care—particularly in underserved areas. As AI continues to evolve, its ethical implementation, seamless integration into clinical workflows, and ongoing evaluation will be critical to ensuring its responsible and impactful use in healthcare.

Looking ahead, the focus is on developing trustworthy and ethical AI systems in medicine. Initiatives like the FUTURE-AI consortium aim to establish guidelines ensuring AI tools are transparent, fair, and aligned with clinical needs. Emphasis is also placed on addressing biases in healthcare data to prevent disparities in AI-driven care. The integration of explainable AI models is anticipated to enhance clinician trust and facilitate informed decision-making, paving the way for more widespread adoption of AI in healthcare.

Abbreviations

AI	Artificial Intelligence
NLP	Natural Language Processing
RPM	Remote Patient Monitoring
ML	Machine Learning
IOT	Internet of Things
WHO	World Health Organization
EHR	Electronic Health Records

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Competing interests
The authors declare no competing interests.

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